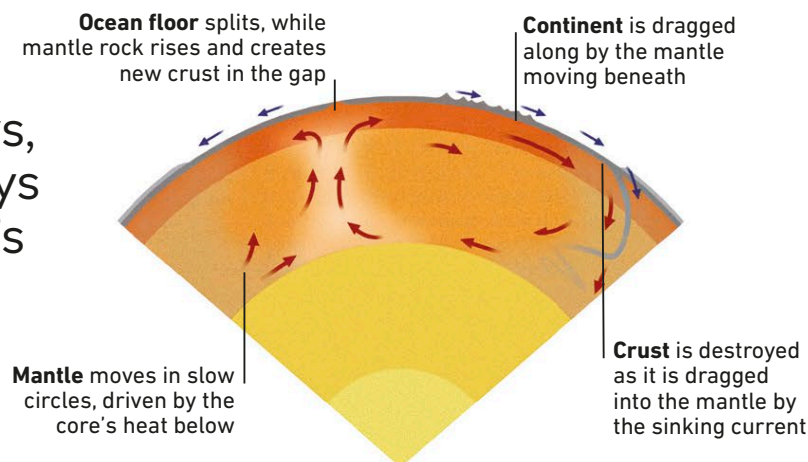


Introduction

Earth is a planet in motion, spinning on its axis as it hurtles through space around the sun. Warmed by the sun's rays, Earth's atmosphere and oceans are always on the move, while heat from the planet's core keeps the hot rock of the interior constantly churning. All of this enables Earth's surface to teem with life.

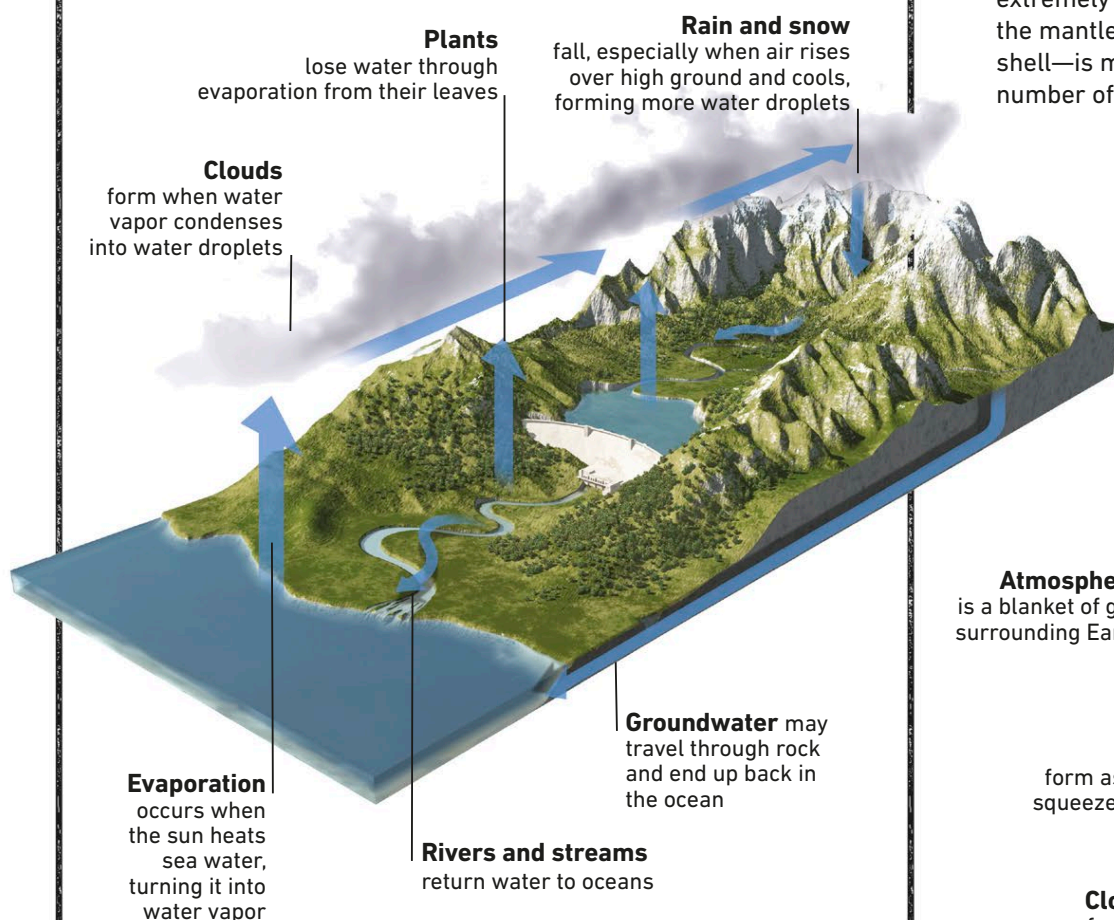
Churning interior

The rocks in the mantle flow in currents that rise, flow sideways, cool, and then sink. These currents can force the plates of Earth's crust apart or pull sections of the crust back down into the mantle.



Water cycle

The sun's heat evaporates sea water, causing it to become water vapor in the air. As it rises and cools, the water vapor condenses into clouds of droplets or ice crystals. As the droplets or crystals grow, they fall as rain or snow. If it falls on land, some runs off the surface to form rivers and lakes, which return water to the oceans. A lot of rain seeps through gaps in the soil and rock. It is called groundwater, and it may stay underground or trickle to the sea. This continuous circulation of water is known as the water cycle.



Earth's structure

If we could take a slice out of Earth, we would see that the planet is made up of layers. At its heart lies a solid inner core, surrounded by a liquid outer core. Both are made mainly of heavy iron. The outer core is enclosed by a deep layer of heavy, very hot, yet solid rock called the mantle. Heat from the core drives currents rising through the mantle that keep the rock moving extremely slowly. The crust—the mantle's cool, hard shell—is made up of a number of rocky plates.

Atmosphere
is a blanket of gas surrounding Earth

Mountains
form as the crust is squeezed and folded

Clouds of water droplets
form huge swirling weather systems in the lower atmosphere